

Tableau Lab Record — Detailed Program Steps

Program 1

1. Create calculated fields on the dataset Sample-Superstore using Tableau.

a. Display absolute value of profit for each Region and Sub-Category.

- Open Tableau and connect to dataset
- Navigate to Analysis, choose Create Calculated Field
- Name the field as Absolute Profit
- Enter the formula: `ABS([Profit])`
- Click OK
- Drag Region and Sub-Category to the Rows shelf
- Drag the Absolute Profit to the Text or Label card

b. Find the sum of maximum and minimum profit for each Region.

- Create a new sheet
- Navigate to Analysis, choose Create Calculated Field
- Name the field as Sum of Max and Min Profit
- Enter the formula: `{FIXED [Region]: MAX([Profit])} + {FIXED [Region]: MIN([Profit])}`
- Click OK
- Drag Region to the Rows shelf
- Drag Sum of Max and Min Profit to text or label card

c. Display the sales information of categories in the form of percentage.

- Create a new sheet
- Navigate to Analysis, choose Create Calculated Field
- Name the field as Sales Percentage
- Enter the formula: `SUM([Sales]) / TOTAL(SUM([Sales]))`
- Click OK
- Drag Category to Rows shelf & Sales Percentage to text card
- Right-click on Sales Percentage, select Format, under Numbers, choose Percentage from the list

d. Display the data for Order Date less than 2021 and Sales between 5000 and 10000.

- Drag Order Date to Rows shelf and select Sales
- Drag Order Date to filters shelf, select Years, then click Next
- Check the years before 2021 and click OK
- Drag Sales to filters shelf, choose Range of values, set the range as: Minimum 5000 and Maximum 10000
- Click OK

e. Convert the first 5 characters of State to Uppercase and plot it against Profit.

- Create a new sheet
- Navigate to Analysis, choose Create Calculated Field
- Name the field as Uppercase State
- Enter the formula: `UPPER(LEFT([State],5))`
- Click OK
- Drag Uppercase State to Columns shelf & Profit to Rows shelf

f. Replace '-' with '(' and add ')' at the end of Customer ID, plot it against Profit.

- Create a new sheet
- Navigate to Analysis, choose Create Calculated Field
- Name the field as Formatted Customer ID
- Enter the formula: `REPLACE([Customer ID], "-", "(" + ")"`
- Click OK and drag Sales to Rows shelf & Formatted Customer ID to Columns shelf

g. Display the Order Date in the format mm/dd/yyyy hh:mm:ss AM/PM.

- Create a new sheet
- Navigate to Analysis, choose Create Calculated Field
- Name the field as Formatted Order Date
- Enter the following formula:
`STR(DATEPART('month',[Order Date])) + "/" + STR(DATEPART('day',[Order Date])) + "/" + STR(DATEPART('year',[Order Date])) + " " + IF DATEPART('hour',[Order Date]) = 0 THEN '12' ELSE STR(DATEPART('hour',[Order Date]) % 12) END + ":" + RIGHT("0"+STR(DATEPART('minute',[Order Date])),2) + ":" + RIGHT("0"+STR(DATEPART('second',[Order Date])),2) + " " + IF DATEPART('hour',[Order Date]) >= 12 THEN 'PM' ELSE 'AM' END`
- Drag Formatted Order Date to Rows shelf & Sales to Columns shelf

Dashboard 2. Create a Dashboard considering at least 4 charts from the above calculated fields.

- Create a New Dashboard
- Drag any 4 charts from the calculated fields
- Arrange them as you see fit

Program 2

1. Create calculated fields on the dataset Amazon.

a. Round-off the Rating to zero decimal places and plot it against User Name.

- Open Tableau and connect to dataset
- Navigate to Analysis, choose Create Calculated Field
- Name the field as Rounded Rating
- Enter the formula: `ROUND([Rating],0)`
- Click OK and drag User Name to Columns shelf and Rounded Rating to Rows shelf

b. Obtain the plot of rating versus first 4 characters of the Product Name.

- Create a New sheet
- Navigate to Analysis, choose Create Calculated Field
- Name the field as F4 Chr of Product Name
- Enter the formula: `LEFT([Product Name],4)`
- Click OK and drag F4 Chr of Product Name to Columns shelf and Rating to Rows shelf

c. Capitalize only the first letter of the Product Name and plot it against any measure.

- Create a new sheet
- Navigate to Analysis, choose Create Calculated Field
- Name the field as Capitalized Product Name
- Enter the formula: `UPPER(LEFT([Product Name],1)) + LOWER(MID([Product Name],2))`
- Click OK & drag Capitalized Product Name to Columns shelf and Rating to Rows shelf

d. Determine if there are null values in the Rating and replace it with zero.

- Create a new sheet
- Navigate to Analysis, choose Create Calculated Field
- Name the field as ReplaceFunction
- Enter the formula: IFNULL([Rating],0)
- Drag or click Product Name, Rating and ReplaceFunction

e. Find the average Sales Price.

- Create a new sheet
- Navigate to Analysis, choose Create Calculated Field
- Name the field as SalePrice and enter the formula: [Actual Price] - [Discounted Price]
- Click OK
- Create another calculated field and name it as Average Sales Price and enter the formula: AVG([SalePrice])
- Click OK and drag Average Sales Price to Rows shelf & label

f. Capitalize the first character of every word under the Product Name & plot it against Discounted Price.

- Create a new sheet
- Drag Capitalized Product Name on Rows shelf
- Drag Discounted Price on Columns shelf
- Navigate to Show Me and select text table

g. Round-off the actual price to the nearest lowest integer and plot it against Product Names starting with 'AG'.

- Create a new sheet
- Navigate to Analysis, choose Create Calculated Field
- Name the field as Rounded Actual Price
- Enter the formula: FLOOR([Actual Price])
- Click OK
- Drag and drop Product Name to filters, select Wildcard, check starts with, enter 'AG' in match value and click OK
- Drag Product Name to Rows shelf and Rounded Actual Price to Columns shelf
- Navigate to Show Me, select horizontal bars

h. Draw a Stacked bar plot depicting the Product ID with series 'B07' against Rating.

- Create a new sheet
- Drag and drop Product ID to filters shelf
- Select Wildcard, check Starts with, enter 'B07' in match value and click OK
- Drag and drop Product ID and Rating
- Navigate to Show Me and select Stacked bars

2. Create a dashboard considering at least 4 charts from the above calculated graphs.

- Create a new dashboard
- Drag & drop any 4 charts from the calculated fields
- Arrange them as you see fit

Program 3

1. Demonstrate the Data Extraction and different types of Joins on the dataset Sample-Superstore using Tableau.

- Open Tableau and connect to dataset
- In the upper-right corner of data source page select Extract option
- Click on sheet tab and Tableau will prompt you to save the extract as a hyper file which is a local copy of your dataset
- Drag first table to the canvas and double click the first table to open the physical layer
- Double click or drag another table to join the canvas
- Click the join icon/figure to configure the join (Inner, Left, Right and Full Outer)

2. Demonstrate the following charts using Sample-Superstore.

a. Dual Axis Chart:

- Drag Sales and Profit to Rows shelf
- On the second graph, click on the drop-down arrow on the Profit pill in the Rows shelf and select Dual Axis
- Right-click on one of the Y-axes and select Synchronize axis to ensure both axes align correctly

b. Motion Chart:

- Create a new sheet
- Drag Sales to Columns shelf & Profit to Rows shelf
- Drag Region to the Detail card in the Marks pane
- Drag Order Date to the Pages shelf
- Press play to watch how the scatter plot evolves over time

c. Gantt chart:

- Create a new sheet
- Select Product ID, Product Name and Order Date
- Navigate to Show Me, select Gantt view

d. Bullet chart:

- Create a new sheet
- Select Category, Profit and Quantity
- Navigate to Show Me, select bullet graphs

e. Bubble chart:

- Create a new sheet
- Select Region, Profit and Sales
- Navigate to Show Me, select packed bubbles

f. Area chart:

- Create a new sheet
- Drag Order Date to Columns shelf & Sales to Rows shelf
- Navigate to Show Me, select area charts (continuous)
- Right-click the Order Date pill in Columns shelf & select month/year depending on time granularity you want to see
- Drag Region to the Color shelf

g. Tree map:

- Create a new sheet
- Select Region, Quantity and Sales

- Navigate to Show Me, select treemaps

h. Heat Map:

- Create a new sheet
- Select State, Profit and Sales
- Navigate to Show Me, select heat maps

3. Create a dashboard considering at least 4 charts from the above charts.

- Create a new dashboard
- Drag any 4 charts from the above charts
- Arrange them as you see fit

Program 4

1. Create Filters and analyze the data for the given dataset Coffee Chain using Tableau.

a. What is the number of sales in the state Nevada for the product Decaf Espresso.

- Open Tableau and connect to dataset
- Drag State to filters shelf and select Nevada & click OK
- Drag Product to filters shelf & select Decaf Espresso & click OK
- Select Product, State and Sales
- Navigate to Show Me, select text tables

b. In California, which is the product with the highest and lowest profit.

- Create a new sheet
- Drag State to filters shelf & select California and click OK
- Select State, Product and Profit
- Navigate to Show Me, select horizontal bars
- Select sort descending
- Drag State to columns

c. What is the contribution of sales in the East market for Decaf.

- Create a new sheet
- Drag market to filters shelf, select East & click OK
- Drag Product to filters shelf, select Decaf & click OK
- Select Product and Sales
- Drag market to the Columns shelf

d. In 2012, what is the contribution of sales in the East market for Decaf.

- Create a new sheet
- Drag Date to filters shelf, select year, check 2012 & click OK
- Drag market to filters shelf, select East, click OK
- Drag Product to filters shelf, select Decaf & click OK
- Drag Product to Rows shelf & Market and Date to Columns shelf

e. What is the average profit for all the products starting with 'C'?

- Create a new sheet
- Drag Product to filters shelf, select Wildcard, check starts with, enter 'C' in match value & click OK

- Drag Product to Rows shelf & Profit to Columns shelf
- Right-click on Profit pill, navigate to Measure & select average
- Drag Profit to text/label card, right-click & navigate to measure and select average

f. In 2013, identify the state with the highest Profit in the West market.

- Create a new sheet
- Drag market to filters shelf, select West & click OK
- Drag Date to filters shelf, select years, check 2013 & click OK
- Select State and Profit
- Navigate to Show Me and select horizontal bars
- Drag market to columns shelf
- Drag Profit to label card

g. Identify the total expenses to sales ratio of the state with the lowest profit.

- Create a new sheet
- Drag State to Rows and Profit to Columns shelf
- Right-click on Profit pill and choose Sort ascending
- Navigate to Analysis, choose Create Calculated Field
- Name the field as Expense to Sales Ratio and enter the formula: $\text{[Total Expenses]} / \text{[Sales]}$ & click OK
- Drag State to filters shelf & select state having lowest profit identified in step c
- Drag Expense to Sales Ratio to Columns shelf

h. In 2013, what is the percentage of total profit for Caffè Mocha falling under Major Market?

- Create a new sheet
- Drag Date to filters shelf, select 2013 and click OK
- Drag Product to filters shelf, select Caffè Mocha and click OK
- Drag market size to filters shelf, select Major market & click OK
- Drag Profit to Rows shelf
- Navigate to Analysis, choose Create Calculated Field
- Name the field as Percentage of Total Profit
- Enter the formula: $\text{SUM}([\text{Profit}]) / \text{TOTAL}(\text{SUM}([\text{Profit}]))$ and click OK
- Drag the Percentage of Total Profit to Columns shelf, right-click, choose Format, under Numbers, select Percentage
- Drag Percentage of Total Profit to Text/Label card

2. Create a dashboard considering at least 4 charts from the above charts.

- Create a new dashboard
- Drag any 4 charts from the above charts
- Arrange them as you see fit

Program 5

1. Analyze the data for the given queries by creating charts for the dataset Coffee Chain.

a. Create a Heat map for Product Type, State and Profit. Identify which state has the lowest profit for Espresso.

- Open Tableau and connect to dataset
- Drag Product Type to filters shelf, select Espresso & click OK
- Drag Product Type to Columns shelf & State to Rows shelf

- Select Profit
- Navigate to Show Me, select Heat maps & sort in ascending

b. Using boxplot, identify which of the Espresso product has the highest distribution of sales.

- Create a new sheet
- Drag Product Type to filters shelf, select Espresso & click OK
- Drag Product to Rows shelf & Sales to Columns shelf
- Navigate to Show Me, select box-and-whisker plots
- Drag Sales to Label card

c. Create a bar chart with Product Type, Product & Profit. Find the Product and Product Type with the highest profit and lowest profit.

- Create a new sheet
- Select Product, Product Type and Profit
- Navigate to Show Me, select horizontal bars
- Sort them in descending order
- Drag Profit to Label card

d. Create a Scatter Plot with State, Sales & Profit. Draw your conclusion.

- Create a new sheet
- Select State, Sales and Profit
- Navigate to Show Me, select scatter plot

e. Identify the state with the highest and lowest profit in the market.

- Create a new sheet
- Drag State to Rows shelf and Profit to Columns shelf
- Navigate to Show Me, select horizontal bars
- Sort Descending
- Drag Profit to Label card

f. What is the contribution of tea (in percentage) to the overall profit?

- Create a new sheet
- Drag Product Type to filters shelf, select Tea & click OK
- Navigate to Analysis, select Create Calculated Field
- Name the field as Tea Profit %
- Enter the formula: $\text{SUM}([\text{Profit}]) / 259543 * 100$
- Click OK
- Drag Tea Profit % to Rows shelf
- Drag Profit to Label card

g. Identify through side-by-side circles, the minimum marketing for the coffee beans Colombian product.

- Create a new sheet
- Drag Product to filters shelf, select Colombian & click OK
- Drag Marketing to the Rows shelf and Market to the Columns shelf
- Drag Product to the Columns shelf
- Navigate to Show Me, select side-by-side circles

h. Using treemap, identify the market with its size having maximum budget sales.

- Create a new sheet
- Select market and Budget Sales
- Navigate to Show Me, select treemaps
- Drag Budget Sales to Label card

2. Create a dashboard considering at least 4 charts from the above charts.

- Create a new dashboard
- Drag any 4 charts from the above charts
- Arrange them as you see fit

Program 6

1. Analyze the data for the given queries by creating charts for the dataset Office Supplies.

a. Find the average units offered by the representative Richard.

- Open Tableau and connect to dataset
- Drag Rep to filters shelf, select Richard & click OK
- Drag Units to Rows shelf, right click, navigate to measure and select average
- Drag Rep to the Label card

b. Find the total units offered by the representative Richard.

- Create a new sheet
- Drag Rep to filters shelf, select Richard and click OK
- Drag Units to Rows shelf and Label card
- Drag Rep to the Label card

c. What is the unit price of all items starting with P?

- Create a new sheet
- Drag Item to filters shelf, select Wildcard, check Starts with, enter 'P' in match value, click apply & OK
- Drag Item to Columns shelf & Unit Price to Rows shelf
- Drag Unit Price to Label card
- Navigate to Show Me, select horizontal bars

d. Find the Unit Price of Binders in the West region.

- Create a new sheet
- Drag Item to filters shelf, select Binder and click OK
- Drag Region to filters shelf, select West and click OK
- Select Item, Region and Unit Price

e. What is the average Unit Price of Pen Set in the East Region?

- Create a new sheet
- Drag Item to filters shelf, select Pen Set, and click OK
- Drag Region to filters shelf, select East, and click OK
- Select Item, Region and Unit Price
- Right-click on Unit Price, navigate to measure, select average

f. Create a Scatter plot depicting the relation between any 3 variables.

- Create a new sheet
- Select Item, Units and Unit Price
- Navigate to Show Me, select scatter plots
- Drag Item to color card & Unit Price to label card

g. Create a highlight table depicting the relation between any 3 variables.

- Create a new sheet
- Select Item, Order Date and Unit Price
- Navigate to Show Me, select highlight tables

h. Create a stacked bar depicting the relation between region, item and units.

- Create a new sheet
- Select Region, Item and Units
- Navigate to Show Me, select stacked bars
- Drag Units to label card
- Drag Region to label card

2. Create a Dashboard considering at least 4 charts from the above charts.

- Create a new dashboard
- Drag any four charts from the above charts
- Arrange them as you see fit